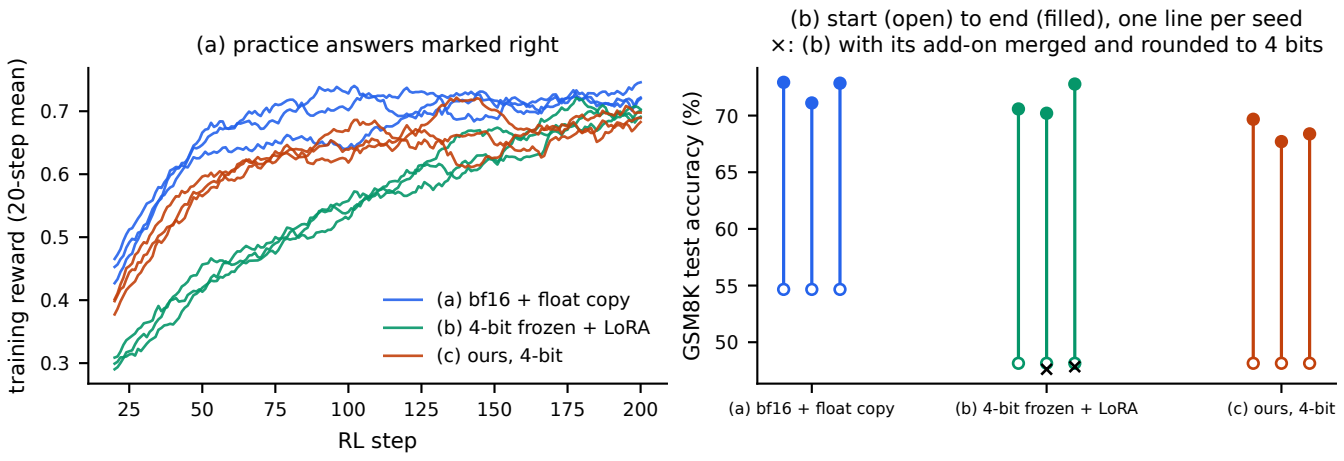
(a) practice answers marked right
training reward (20-step mean)
0.3
0.4
0.5
0.6
0.7
25
50
75
100
125
150
175
200
RL step
(a) bf16 + float copy
(b) 4-bit frozen + LoRA
(c) ours, 4-bit
(b) start (open) to end (filled), one line per seed
×: (b) with its add-on merged and rounded to 4 bits
GSM8K test accuracy (%)
50
55
60
65
70
(a) bf16 + float copy
(b) 4-bit frozen + LoRA
(c) ours, 4-bit